Elasticity and Its Applications

Measuring responsiveness → demand and revenue → supply → tax incidence

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1. The meaning of elasticity

Supply and demand tell us the *direction* of a response. When a price rises, quantity demanded usually falls. But how much does it fall? Does it fall by 2% or 20%? That difference matters for a shop considering a price increase, a firm expanding production, or a government imposing a tax. *Elasticity* measures the size of a response.

> **DEFINITION Elasticity**
>
> **Elasticity** measures how responsive one variable is to a change in another. It is calculated as a **percentage change divided by a percentage change**. An elasticity of 2 means the measured variable changes proportionately twice as much as the variable causing the change (for small changes, holding other influences fixed).

Why percentages? Suppose one business sells 500 tickets and another sells 5,000. Losing 100 tickets is a 20% decline at the first business but only a 2% decline at the second. The same absolute change has a very different meaning. Percentages put the two markets on a common scale.

> **KEY RESULT Why elasticity is more useful than a raw slope**
>
> A slope compares a change in units with a change in dollars. It depends on how we measure the good and the price. **Elasticity compares percentage changes**, so it is a unit-free number. It can compare responsiveness across different markets.

2. Own-price elasticity of demand

> **DEFINITION Own-price elasticity of demand**
>
> The **own-price elasticity of demand**, E_d, is the percentage change in quantity demanded divided by the percentage change in the *same good's* price, with other influences on demand unchanged:
>
> $$E_d = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}.$$
>
> By the law of demand the signed value is normally negative. In this handout, $|E_d|$ means its **absolute value** (ignore the minus sign) when classifying demand as elastic or inelastic.

For example, a 10% price increase accompanied by a 15% reduction in quantity demanded gives $E_d = -15/10 = -1.5$ and $|E_d| = 1.5$. The quantity response is greater, in percentage terms, than the price change. This says more than merely observing that demand slopes down.

The midpoint method: the reliable calculation for two observations

Consider a local recreation center's *single-entry day passes*. We will use the same hypothetical demand curve for the next several examples. At a price of \$4 the center sells 120 passes per month; at \$5 it sells 100. These are two points on the *same demand curve*, with income and other factors held constant.

Worked example: Why ordinary percentage changes give conflicting answers

From \$4 to \$5, price rises $1/4 = 25\%$ and sales fall $20/120 = 16.67\%$. The absolute elasticity would be $16.67/25 = 0.667$. Going backward, price falls $1/5 = 20\%$ and sales rise $20/100 = 20\%$, giving $20/20 = 1.00$. The same pair of points should not yield different answers merely because we reverse direction.

> **DEFINITION Midpoint (arc) elasticity**
>
> For any two observations, use the *average* of the endpoints as the base for each percentage change:
>
> $$\boxed{|E_d| = \left| \frac{(Q_2 - Q_1)/\big[(Q_1 + Q_2)/2\big]}{(P_2 - P_1)/\big[(P_1 + P_2)/2\big]} \right|.}$$
>
> P_1, P_2 are the two prices; Q_1, Q_2 are the associated quantities. The symbol $|\ |$ means take the absolute value. The result is identical in either direction.

Worked example: Calculate the midpoint elasticity, one step at a time

The average quantity is $(120 + 100)/2 = 110$ passes. The quantity change is $100 - 120 = -20$. Its midpoint percentage change is $-20/110 = -18.18\%$. The average price is $(4 + 5)/2 = \$4.50$. The price change is \$1, or $1/4.5 = 22.22\%$ of that average. Therefore

$$|E_d| = \left| \frac{-20/110}{1/4.5} \right| = \frac{18.18\%}{22.22\%} = \boxed{0.818 \text{ (approximately)}}.$$

Interpretation. A 1% change in price corresponds to approximately a 0.82% change in quantity in the opposite direction, over this interval. Demand is inelastic here. Using averages eliminates the arbitrary choice of starting point.

The straight-line shortcut: slope times a ratio

Suppose the demand schedule is represented by the straight-line equation

$$Q_d = 200 - 20P,$$

where P is dollars per pass and Q_d is passes per month. Read the equation literally: if price rises by \$1, buyers demand 20 fewer passes. The *change in quantity per dollar of price* is $\Delta Q/\Delta P = -20$.

KEY RESULT Slope times the price-to-quantity ratio

For a straight-line demand equation $Q_d = a - bP$ (with $b > 0$), the **point** elasticity is $|E_d| = b(P/Q)$. To calculate the **midpoint** elasticity between any two points on that line, use

$$|E_d| = b\frac{\text{average price}}{\text{average quantity}} = b\frac{\overline{P}}{\overline{Q}}.$$

Here $\overline{P}$ and $\overline{Q}$ mean the averages of the two prices and two quantities. This is the same midpoint formula written more compactly, not a different method.

In our example, $b = 20$, average price is \$4.50, and average quantity is 110. Thus $20(4.5/110) = 0.818$, exactly the result above. **Important:** In a conventional economics graph with price on the vertical axis, the drawn slope is $\Delta P/\Delta Q$, the *reciprocal* of $\Delta Q/\Delta P$. Use the coefficient of P in the equation $Q = a - bP$, not the graphical rise-over-run, in this shortcut.

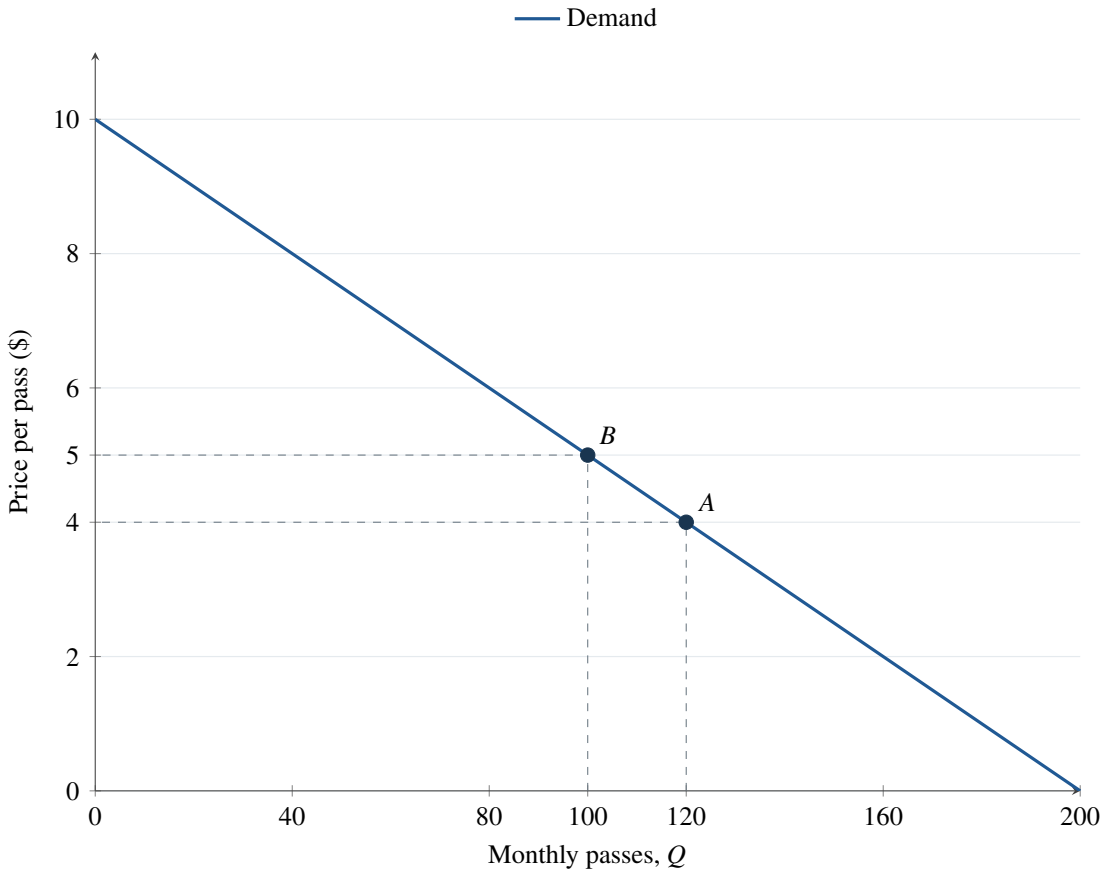


Figure 1. Two points on the same demand curve. The price rises from \$4 to \$5 and monthly quantity demanded falls from 120 to 100. The midpoint calculation uses both endpoints.

Read the graph. Start at point A (120 passes at \$4); move to B (100 at \$5). The fall in sales is 20 passes. The graph shows *where* the quantities come from; the midpoint formula measures *how responsive* buyers are.

3. Elastic, inelastic, and unit-elastic demand

DEFINITION Three classifications

Elastic demand: $|E_d| > 1$; quantity changes by a larger percentage than price.
Inelastic demand: $0 \leq |E_d| < 1$; quantity changes by a smaller percentage than price.
Unit-elastic demand: $|E_d| = 1$; quantity and price change by equal percentages in magnitude.

KEY RESULT What the comparison to one means

Always put the **percentage change in quantity over the percentage change in price**. If that ratio has absolute value 1.7, the demand response is proportionately larger: elastic. If it is 0.6, the response is smaller: inelastic. The dividing line is **one**, not zero.

A straight line does not have constant elasticity

Keep the recreation center's $Q_d = 200 - 20P$. Its slope is always -20 passes per dollar. But the percentage effect of 20 passes depends on the initial quantity, and the percentage effect of one dollar depends on the initial price. Therefore elasticity changes along the same straight line.

Price P	Quantity Q	Calculate $20P/Q$	$\lvert E_d \rvert$	Classification
\$8	40	$20(8/40)$	4.00	Elastic
\$5	100	$20(5/100)$	1.00	Unit elastic
\$2	160	$20(2/160)$	0.25	Inelastic

At the top of this demand curve, a price change is large relative to a small quantity, so demand is elastic. At the bottom, price is low relative to a large quantity, so demand is inelastic. At the exact midpoint of a downward-sloping straight-line demand curve, demand is unit elastic.

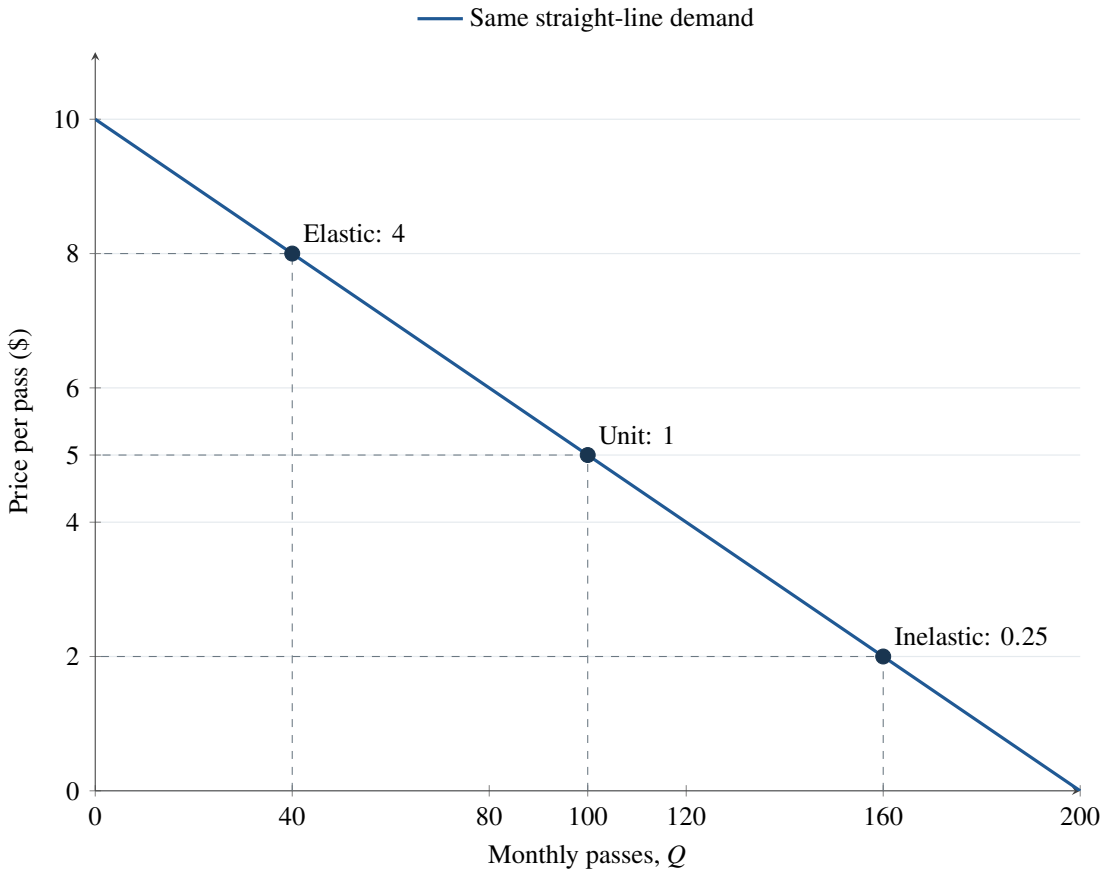


Figure 2. Constant slope, changing elasticity. The upper part is elastic, the midpoint unit elastic, and the lower part inelastic. The labels are calculated at the marked *points*.

Read the graph. The price falls as we move rightward along the line. Compare the three marked points. Although the line does not bend, the elasticity falls from 4 to 1 to 0.25. **Slope is not elasticity.**

4. The determinants of demand elasticity

What makes buyers respond strongly to a price change? Ask a practical question: *how easily can they reduce their purchases, switch to something else, or wait?*

DEFINITION Five important determinants

1. Close substitutes. More easy substitutes usually make demand more elastic.

2. Necessity versus luxury. Purchases regarded as essential tend to be less elastic; optional purchases tend to be more elastic.

3. Market definition. Demand for one narrowly defined product generally has more substitutes than demand for a broad product category.

4. Share of the buyer's budget. Buyers usually respond more when a purchase absorbs a substantial share of income.

5. Time to adjust. Demand generally becomes more elastic over a longer period because buyers can change habits and find alternatives.

Suppose the price of a specific phone-case design rises. A buyer can choose another design, so the demand for that particular case may be quite elastic. But when *all* protective cases rise in price, switching designs no longer escapes the higher category price. The demand for cases as a whole may be less elastic. This is the *market-definition* effect, not an inconsistency.

Similarly, someone who must commute next week may buy nearly the same amount of transportation despite a price increase. Over a year, that person might reorganize travel or move closer to work. Time provides alternatives. None of these determinants assigns one fixed elasticity to a good in all circumstances; the market, the buyers, the starting price, and the time period matter.

KEY RESULT Predict elasticity before calculating it

When comparing two products, identify what makes it easier or harder to change purchases. *More substitutes, a less essential purchase, a narrower market, a bigger budget share, or more adjustment time* generally makes demand more elastic, other things equal. These are tendencies, not numerical laws.

5. Perfectly elastic and perfectly inelastic curves

DEFINITION The two limiting cases for demand

Perfectly inelastic demand: $|E_d| = 0$. Quantity demanded does not change when price changes; its curve is **vertical**.

Perfectly elastic demand: elasticity is unbounded (often written ∞). The curve is **horizontal**: at its specified price, buyers are willing to buy some range of quantities; a price even slightly above it eliminates demand in the idealized model.

Perfectly inelastic: quantity is fixed

Imagine a city has exactly 80 prepaid admission slots already assigned for a one-off event, and the people holding them must each attend regardless of small changes in the admission price. In this *hypothetical model*, the number attending remains 80. The curve is vertical. Actual goods rarely satisfy this assumption at *every* price; the diagram depicts an extreme benchmark, not a factual claim about real behavior.

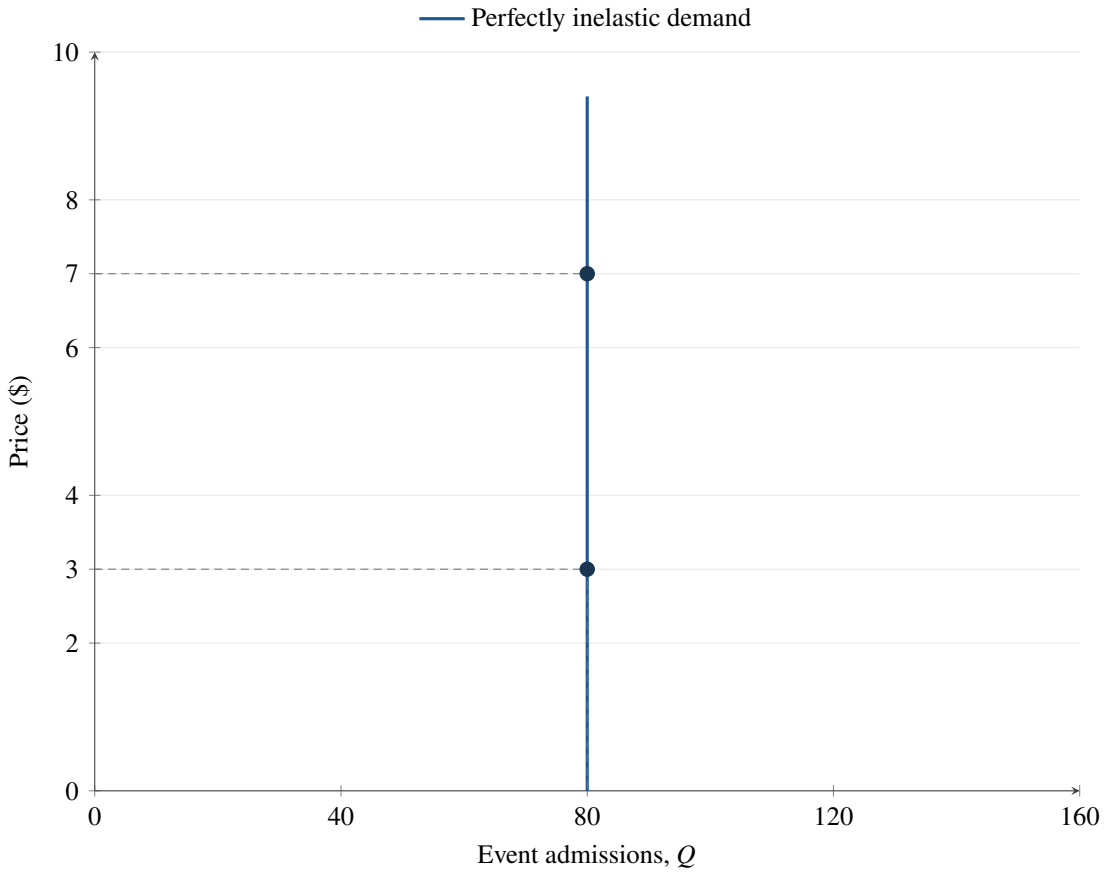


Figure 3. Perfectly inelastic demand. Price changes between the two marked levels, but quantity stays at 80.

Read the graph. Both marked points have the same horizontal coordinate, $Q = 80$. The price changes; quantity does not. Thus the numerator of the elasticity formula is zero and $|E_d| = 0$.

Perfectly elastic: a given price

Imagine a small seller offering a completely standardized digital voucher, indistinguishable from vouchers available from many competing sellers. In the price-taking benchmark, the individual seller cannot charge above the prevailing price because buyers can purchase identical vouchers elsewhere. This is the demand *facing that one seller*, not necessarily the demand for all vouchers in the economy.

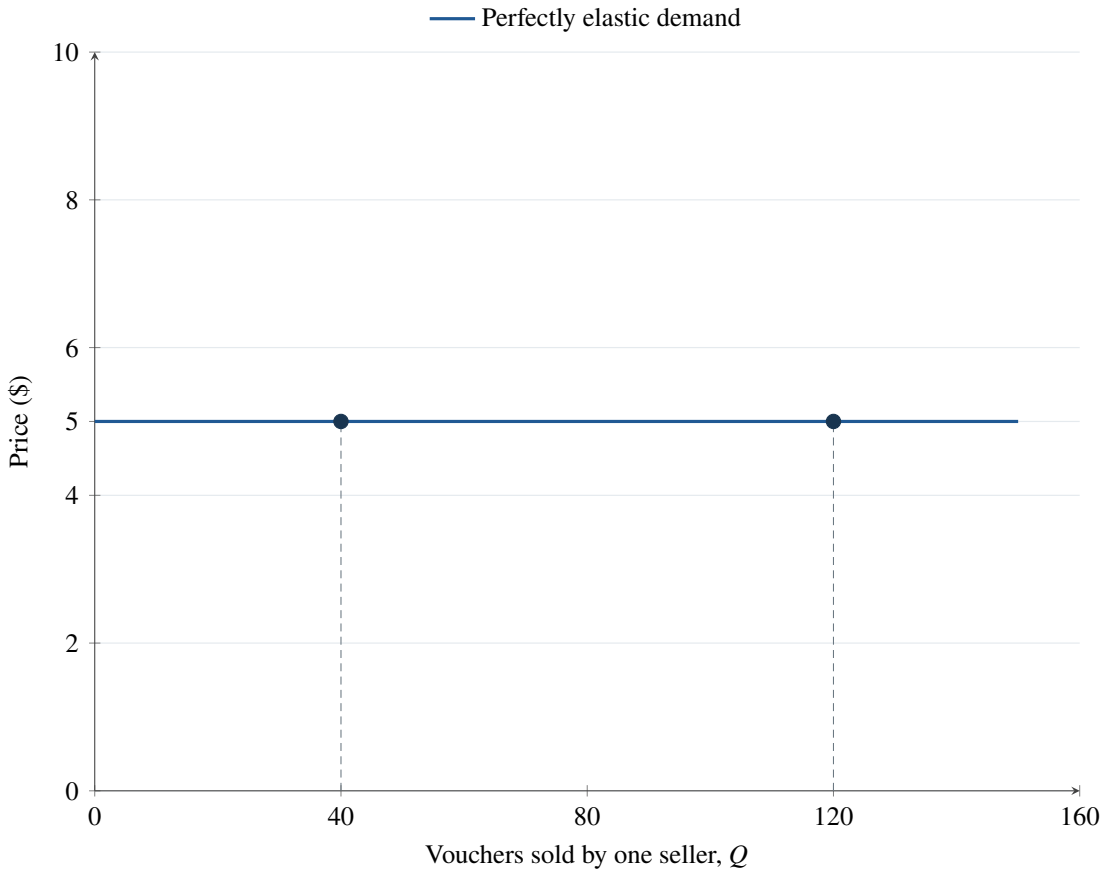


Figure 4. Perfectly elastic demand. The two marked quantities are possible at the same price. A seller pricing above $5 loses customers to identical alternatives in the idealized model.

Read the graph. Price remains at $5 while quantity can differ. A horizontal curve means buyers are *extremely* price sensitive. Do not confuse the horizontal demand curve facing one price-taking seller with the normally downward-sloping *market* demand curve.

6. Elasticity and total revenue

> **DEFINITION Total revenue (and total expenditure)**
>
> **Total revenue**, TR, is the money sellers receive from selling Q units at price P: $\boxed{TR = P \times Q}$. The same amount is buyers' total expenditure on those units. On a demand graph, it is the area of a rectangle whose height is P and width is Q.

If a seller raises the price, each remaining sale brings in more money, but fewer units sell. *Which effect is larger?* Elasticity settles the question. When demand is inelastic the price effect dominates; when elastic, the lost sales dominate.

KEY RESULT Price changes and revenue: the rule to know

Demand at the relevant point or arc	Price increases	Price decreases
Elastic ($\lvert E_d \rvert > 1$)	Revenue falls	Revenue rises
Inelastic ($\lvert E_d \rvert < 1$)	Revenue rises	Revenue falls
Unit elastic ($\lvert E_d \rvert = 1$)	No first-order change	No first-order change

For *finite changes measured with midpoint (arc) elasticity*, an arc elasticity *exactly* equal to one means that revenue is equal at the two endpoints. The phrase "no first-order change" applies to a *small* change at a point where elasticity equals one. These are distinct statements.

Worked example: Revenue on the same recreation-center demand curve

Using $Q = 200 - 20P$, multiply price by the associated quantity. All quantities below are passes per month.

Price	Passes	Revenue $P \times Q$	Point $\lvert E_d \rvert = 20P/Q$	Region
\$2	160	\$320	0.25	Inelastic
\$4	120	\$480	0.67	Inelastic
\$5	100	\$500	1.00	Unit elastic
\$6	80	\$480	1.50	Elastic
\$8	40	\$320	4.00	Elastic

At \$2, raising price moves us through the inelastic part of demand: revenue rises. Above \$5, raising price moves through the elastic part: revenue falls. Revenue reaches its maximum of \$500 at \$5, where *point* elasticity equals one.

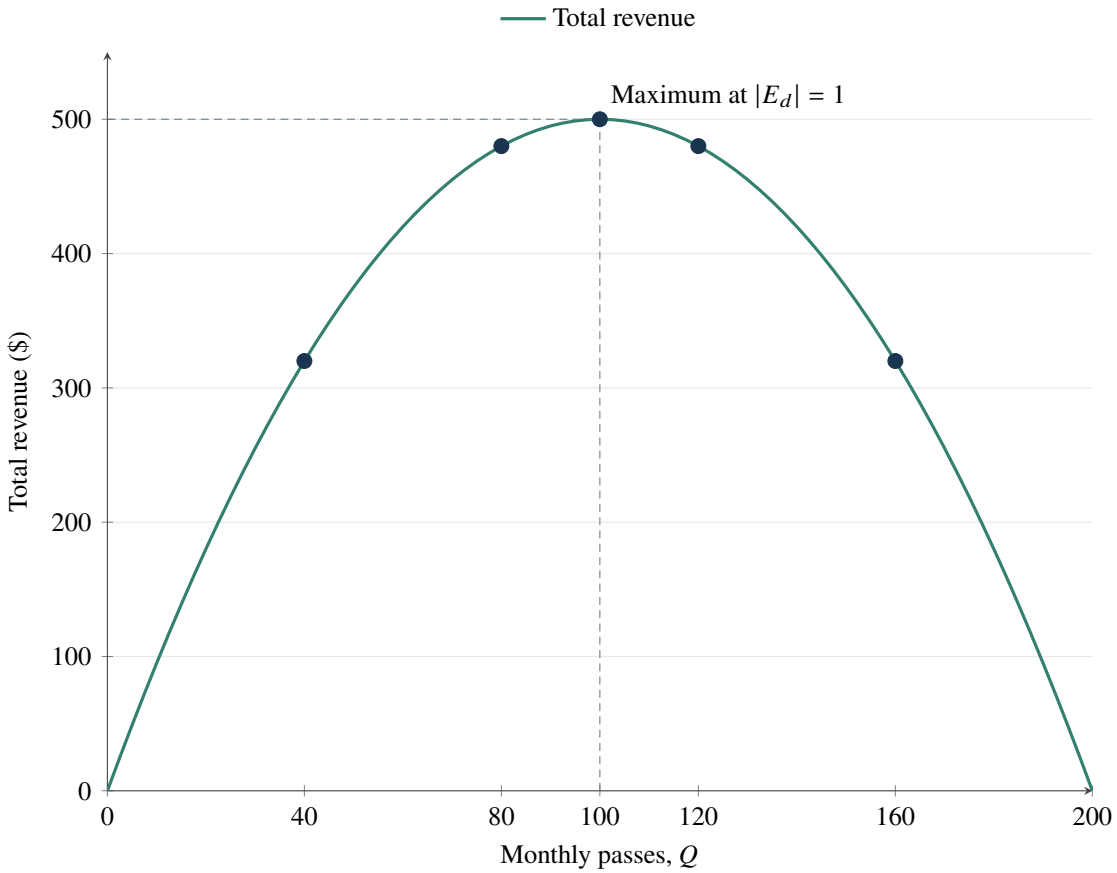


Figure 5. Revenue rises and then falls as sales expand along a straight-line demand curve. It reaches $500 when the price is $5 and the quantity is 100.

Read the graph. Quantity rises from left to right, which means price is falling along the demand curve. On the left, lowering a high price increases revenue because demand is elastic. Beyond 100 passes, lowering price reduces revenue because demand is inelastic. The top is where point elasticity is one.

Worked example: Check a finite price change using midpoint elasticity

Compare \$4 and 120 passes (TR = \$480) with \$6 and 80 passes (TR = \$480). The average price is \$5, average quantity is 100, and the absolute midpoint changes are $2/5 = 40\%$ in price and $40/100 = 40\%$ in quantity. Arc elasticity is exactly one, and the revenues at the endpoints are indeed identical. This does *not* mean the straight-line demand curve is unit elastic everywhere between those prices.

7. The elasticity of supply

So far elasticity has described buyers. Sellers also respond to changes in price. A price rise may induce an immediate increase in production when there is unused equipment, or almost no increase when a facility is at capacity.

DEFINITION Own-price elasticity of supply

The **price elasticity of supply**, E_s, is the percentage change in quantity *supplied* divided by the percentage change in the good's price:

$$\boxed{E_s = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}.}$$

With an upward-sloping supply curve, E_s is normally nonnegative. The *same midpoint formula* applies, replacing quantity demanded with quantity supplied.

Worked example: A poster-printing workshop increases output

Suppose a local printer supplies 80 event posters each week at \$20 per poster and 110 at \$25. Average quantity is $(80 + 110)/2 = 95$. Average price is $(20 + 25)/2 = 22.5$. Consequently

$$E_s = \frac{(110 - 80)/95}{(25 - 20)/22.5} = \frac{30/95}{5/22.5} \approx \boxed{1.42}.$$

Interpretation. Supply is elastic over this interval: output rises proportionately more than price. The workshop might run a longer shift or use idle printers. That is a hypothetical explanation of the example, not a statement about all printing firms.

KEY RESULT Classifying and explaining supply elasticity

$E_s > 1$: elastic supply; $E_s < 1$: inelastic supply; $E_s = 1$: unit elastic. A **vertical** supply curve has $E_s = 0$ (fixed quantity); a **horizontal** supply curve is the perfectly elastic limiting case. Supply is generally more elastic when firms can adjust inputs, have spare capacity, and have *more time* to expand or enter the market.

For example, a sports venue cannot add seats for tonight's sold-out event: the short-run quantity of seats is fixed. Over several years, a venue operator might enlarge the building or competitors could open new venues. That makes a larger supply response possible in the long run. Another important distinction: a straight-line supply curve *through the origin*, $Q_s = bP$, has $E_s = 1$ at every positive point, because quantities and prices move in the same proportion. A straight supply curve that does *not* pass through the origin need not have constant elasticity.

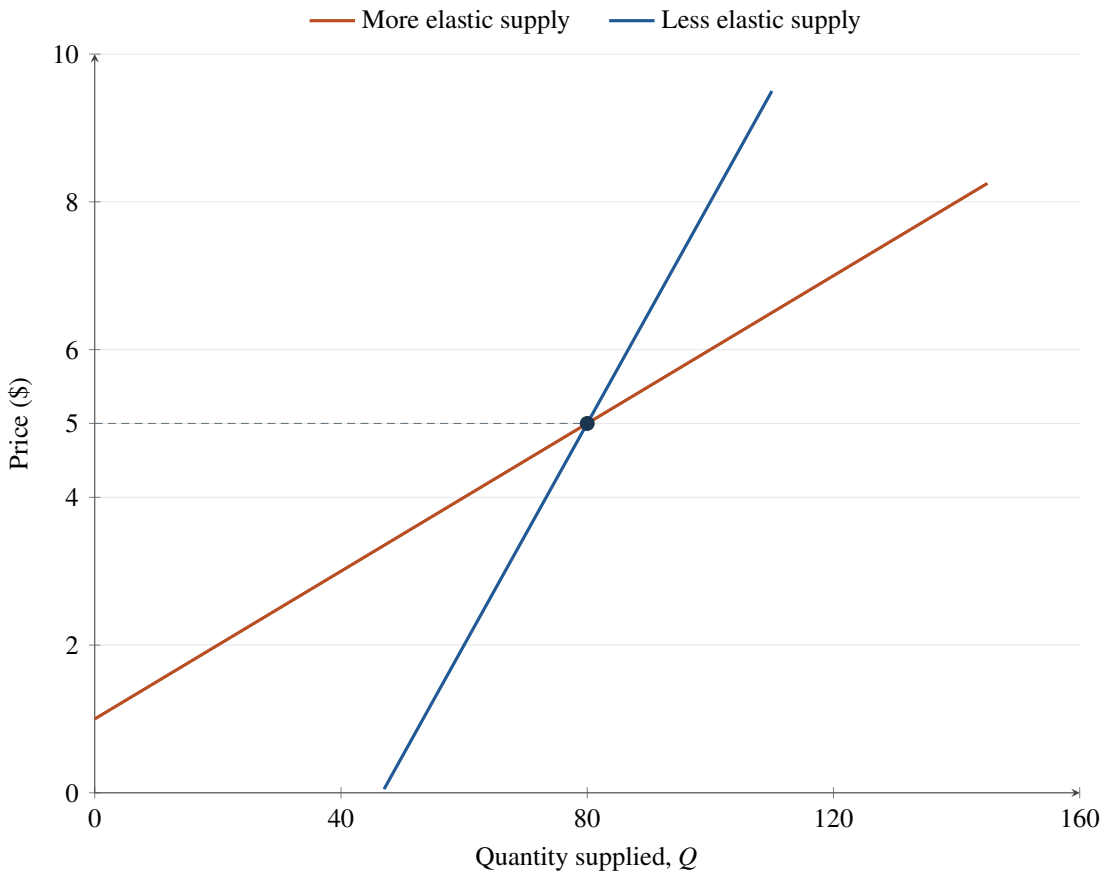


Figure 6. The two curves pass through the same point (80 units at \$5). The flatter orange curve is more elastic at that point: $E_s = 1.25$ versus approximately 0.42 for the blue curve.

Read the graph. Near the common point, the same small price change changes quantity more along orange supply. Both curves use the same axes and price–quantity point, so their different steepness really indicates different elasticities. For an exact number, still calculate the percentage-change ratio.

8. Tax incidence: who really bears a tax?

A government can require sellers to remit a tax, but that does not tell us who ultimately loses purchasing power or income. Prices adjust. Buyers may pay more, sellers may receive less, and fewer units may be traded. *Tax incidence* measures how the burden is divided.

DEFINITION Tax incidence and the tax wedge

Tax incidence is the division of the economic burden of a tax between buyers and sellers. For a tax of t dollars per unit, let P_B be the price *paid by buyers including tax* and P_S the price *received by sellers after tax*. The **tax wedge** is

$$\boxed{P_B - P_S = t.}$$

Compare both prices with the original no-tax price P_0 to measure the burden: buyers bear $P_B - P_0$ per unit and sellers bear $P_0 - P_S$ per unit. The two shares add to t.

KEY RESULT The central tax-incidence rule

The relatively less elastic side of the market bears more of the per-unit tax. If demand is less elastic than supply, buyers bear more. If supply is less elastic than demand, sellers bear more. The result follows because the side with fewer alternatives can reduce its market participation less easily when the price moves against it. Under the standard competitive-market assumptions, changing which side *legally remits* the tax does not change its economic incidence.

Case A: buyers have fewer alternatives

Suppose a market for small repair supplies has a no-tax equilibrium price of \$20 and sells 60 units. For this *hypothetical* example, the demand and supply curves are

$$\text{Buyer willingness to pay: } P_D = 50 - \frac{Q}{2}, \qquad \text{Seller required price: } P_S = 5 + \frac{Q}{4}.$$

At 60 units both sides agree on \$20. Now impose a \$6 per-unit tax. At the new equilibrium of 52 units, buyers will pay $50 - 52/2 = \$24$ and sellers will receive $5 + 52/4 = \$18$. The wedge is $24 - 18 = \$6$. Compared with \$20 before the tax, buyers pay \$4 more and sellers receive \$2 less. Thus buyers bear two-thirds of the tax.

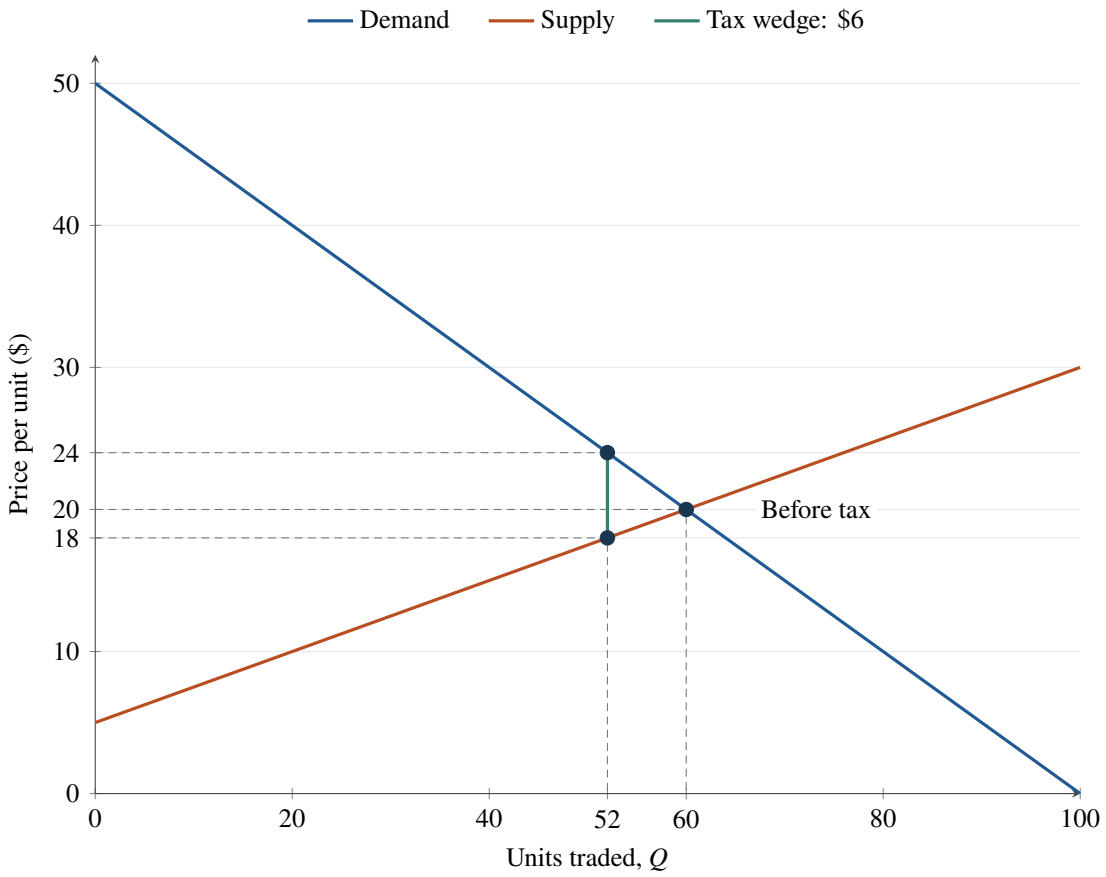


Figure 7. Tax incidence when demand is less elastic than supply. A \$6 tax reduces sales from 60 to 52; buyers pay \$24 and sellers net \$18. Buyers bear \$4, sellers \$2.

Read the graph. The original intersection is at $(Q, P) = (60, 20)$. At the new quantity, draw a vertical line from

supply ($18) to demand ($24). That vertical distance is the $6 tax. Compare each endpoint with the old $20 price: the upper gap is buyers' burden, the lower gap is sellers' burden.

Case B: sellers have fewer alternatives

Now imagine a different market with the *same* initial equilibrium (60 units at $20), but more responsive buyers and less responsive sellers:

$$P_D = 35 - \frac{Q}{4}, \qquad P_S = -10 + \frac{Q}{2}.$$

(The second formula is relevant here only for the positive-price range being studied.) A $6 tax again reduces sales to 52. Buyers pay $35 - 52/4 = \$22$; sellers net $-10 + 52/2 = \$16$. Buyers bear only $2 of the tax and sellers bear $4. Relative elasticities have reversed the burden, even though the tax is the same size.

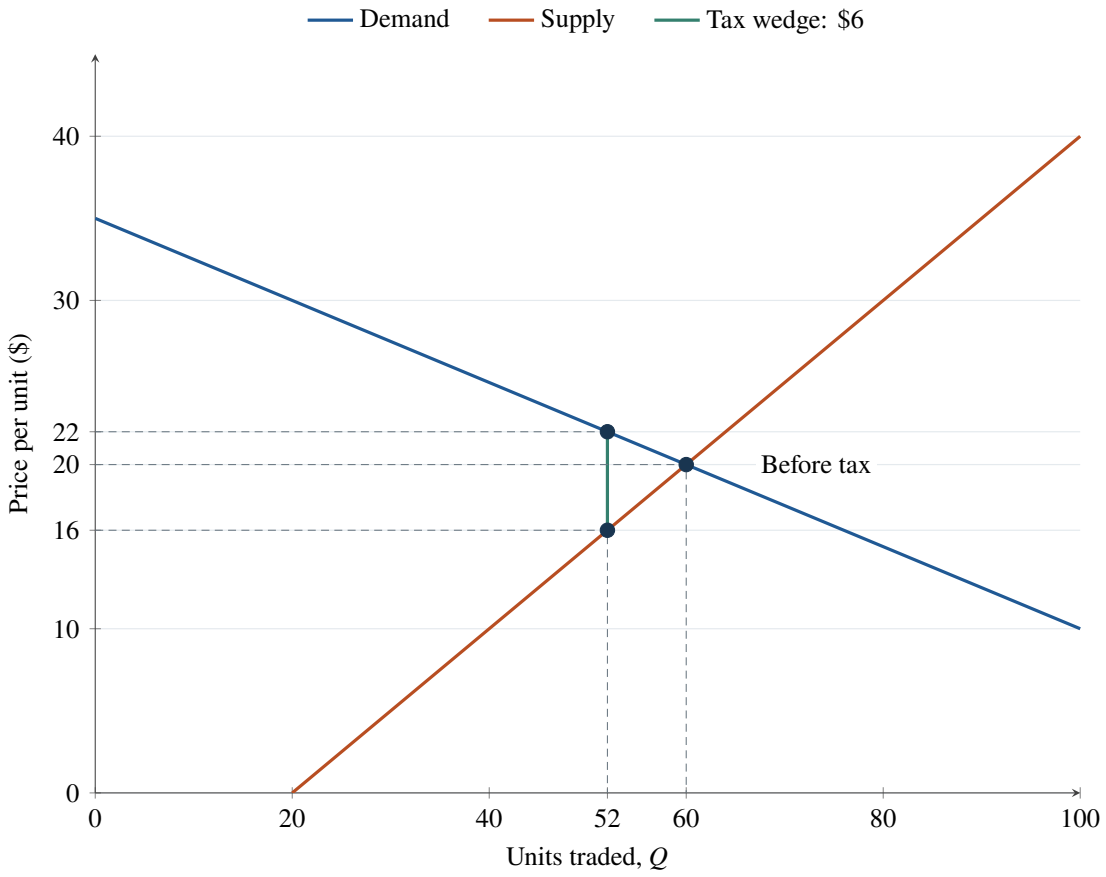


Figure 8. Tax incidence when supply is less elastic than demand. Buyers now pay $22 and sellers net $16; sellers bear $4 of the $6 tax.

Read the graph. The same vertical tax wedge now lies mostly *below* the original $20 price. That is the seller's larger burden. The tax is collected on the 52 units still traded, so government revenue is $52 \times \$6 = \$312$ in either example. The eight units no longer traded do not generate tax revenue.

Why elasticity, not the tax law, determines the burden

In Case A the no-tax point elasticities (in absolute value) are $|E_d| = 2(20/60) = 2/3$ and $E_s = 4(20/60) = 4/3$: demand is less elastic, so buyers bear more. In Case B they reverse: demand is $4(20/60) = 4/3$ and supply is $2(20/60) = 2/3$: sellers bear more. In both cases the side with *less* flexibility bears *more* of the tax. A tax collected from sellers can still raise the price buyers pay; a tax collected from buyers can still reduce the price sellers receive.

KEY RESULT A reliable tax-incidence procedure for examinations

Find the original equilibrium price P_0. Identify whether demand or supply is less elastic. Predict which side bears more. After a tax, find P_B and P_S and check that $P_B - P_S = t$. Finally compare P_B and P_S with P_0. **Who remits the tax is not the same as who bears its burden.**

9. Two other demand elasticities (briefly)

The chapter also introduces elasticities with respect to *income* and the *price of another good*. They use the same percentage-over-percentage idea, but unlike own-price elasticity, their *signs* carry important information.

DEFINITION Income elasticity of demand

$$E_I = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}.$$

Positive E_I: a **normal good** (demand rises with income). Negative E_I: an **inferior good** (demand falls with income). For positive elasticities, a value greater than one means demand rises proportionately faster than income; a value between zero and one means it rises more slowly.

Worked example: Read an income elasticity

Suppose income increases by 10% and a household's purchases of a particular optional service rise by 15%. Then $E_I = 15/10 = +1.5$, so this is a normal good over the stated range. If purchases of a cheaper alternative fall by 3% instead, its income elasticity is $-3/10 = -0.3$, marking it as inferior over that range.

DEFINITION Cross-price elasticity of demand

$$E_{XY} = \frac{\%\text{ change in quantity demanded of good } X}{\%\text{ change in the price of good } Y}.$$

$E_{XY} > 0$ usually means **substitutes**: a higher price of Y raises demand for X. $E_{XY} < 0$ usually means **complements**: a higher price of Y reduces demand for X. Keep the signs.

Worked example: Identify substitutes and complements from the sign

Suppose the price of reusable water bottles increases 10%, while demand for competing insulated mugs rises 5%. Their cross-price elasticity is $+5/10 = +0.5$: substitutes. If purchases of bottle-cleaning brushes instead fall 4%, the relevant cross-price elasticity is $-4/10 = -0.4$: complements. Which good goes on top matters: specify whose *quantity* is changing and whose *price* caused it.

10. Applying elasticity and checking your reasoning

Worked example: A productivity improvement can reduce sellers' total revenue

Suppose a community print cooperative can now produce more simple notebooks at each possible price. Supply shifts right. Initially 100 notebooks sell for \$10 each, so total revenue is \$1,000. After the improvement, 120 sell for \$7 each, so revenue is \$840. The increase in quantity did not compensate for the price decline. On the demand curve between those two outcomes, midpoint price elasticity is

$$|E_d| = \frac{20/110}{3/8.5} \approx 0.515 < 1.$$

Demand is inelastic over this interval. That explains *why* revenue falls when supply expands: quantity sold rises proportionately less than price falls. This says nothing by itself about profits, because costs may also have changed.

> **KEY RESULT What you should be able to do without calculus**
>
> **Recognize** the key concepts; **calculate** own-price demand and supply elasticities with the midpoint method; **use** $b\overline{P}/\overline{Q}$ for a straight-line demand equation; **distinguish** slope from elasticity and point from arc elasticity; **interpret** vertical and horizontal curves; **predict** how a price change affects revenue; and **explain** which side bears more of a tax by comparing relative elasticities. Use income and cross-price elasticity signs to identify normal/inferior goods and substitutes/complements.

Avoid the recurring mistakes

Do not invert the elasticity ratio: it is *percentage quantity change divided by percentage price change*. Do not treat a straight-line slope as a constant elasticity. Do not replace the midpoint denominator with just the starting value. Do not say a price increase shifts demand: with other influences fixed, it moves *along* demand. Do not equate a legal tax payment with the economic burden. And do not infer the *exact* value of elasticity from how steep an unlabeled graph looks.